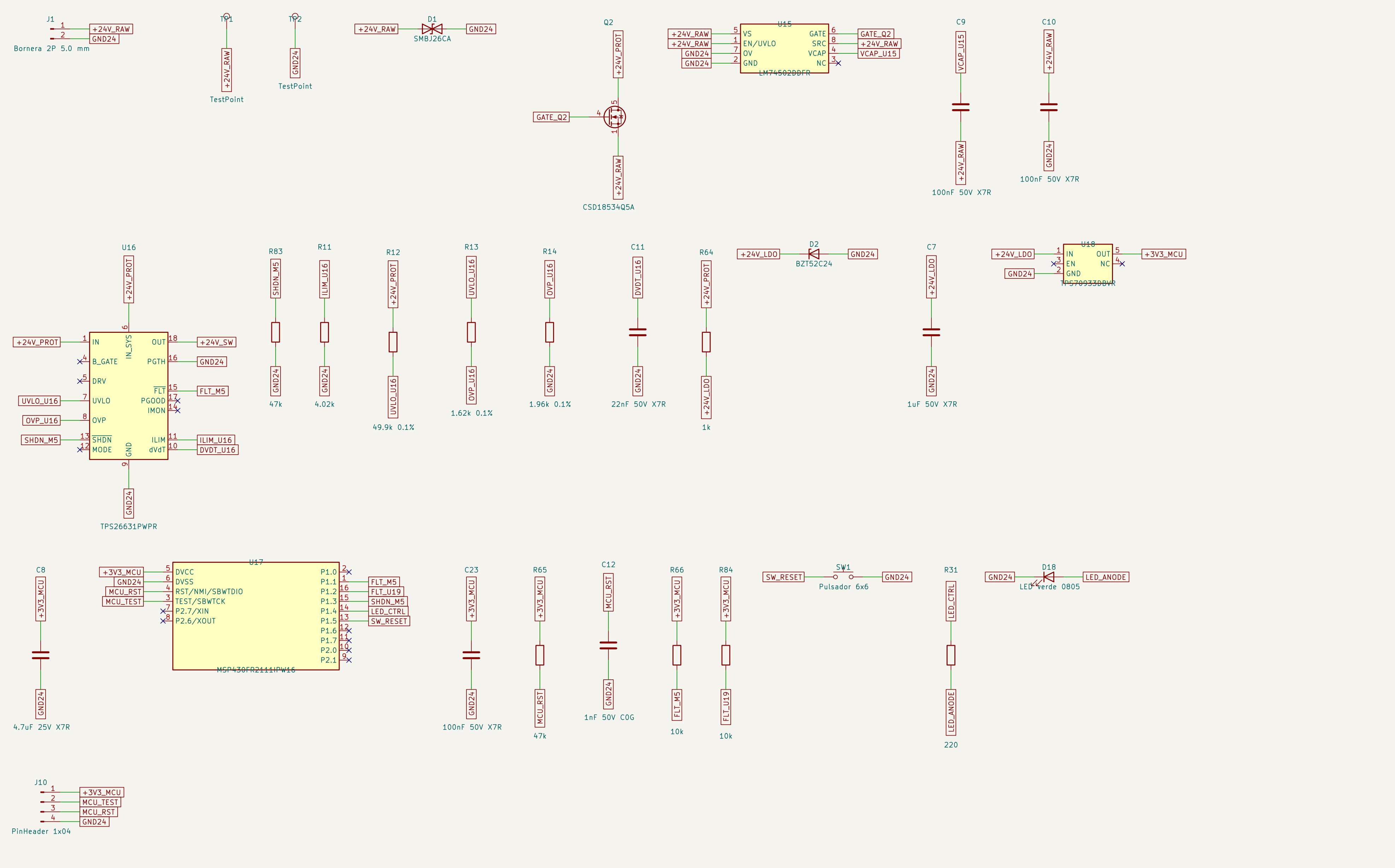
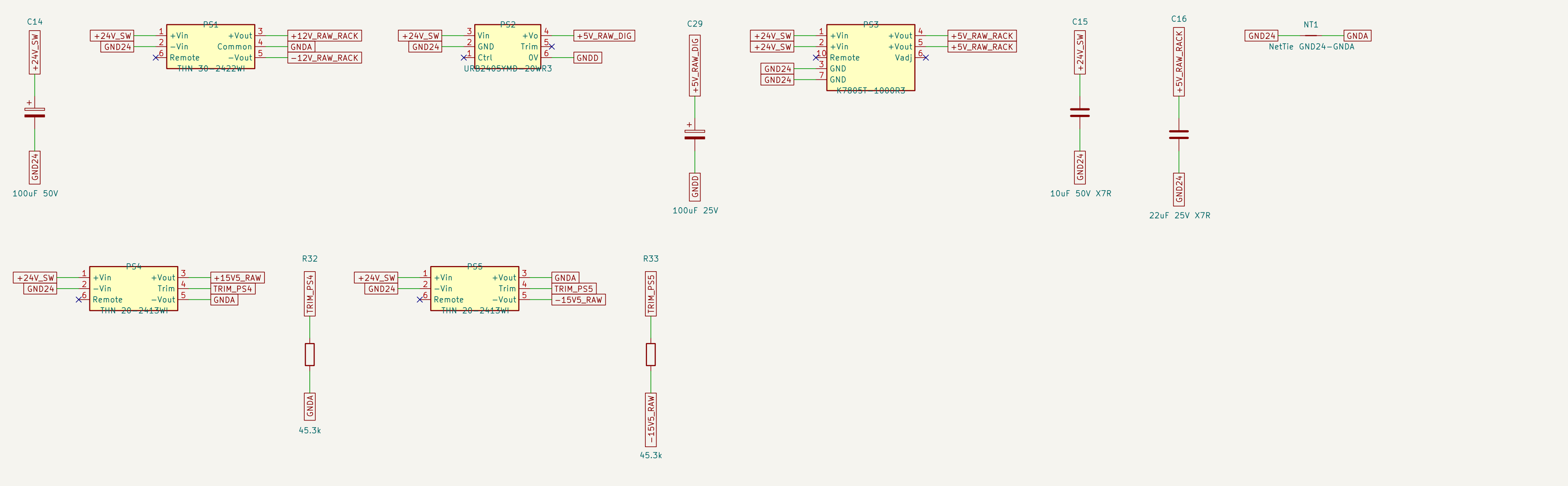
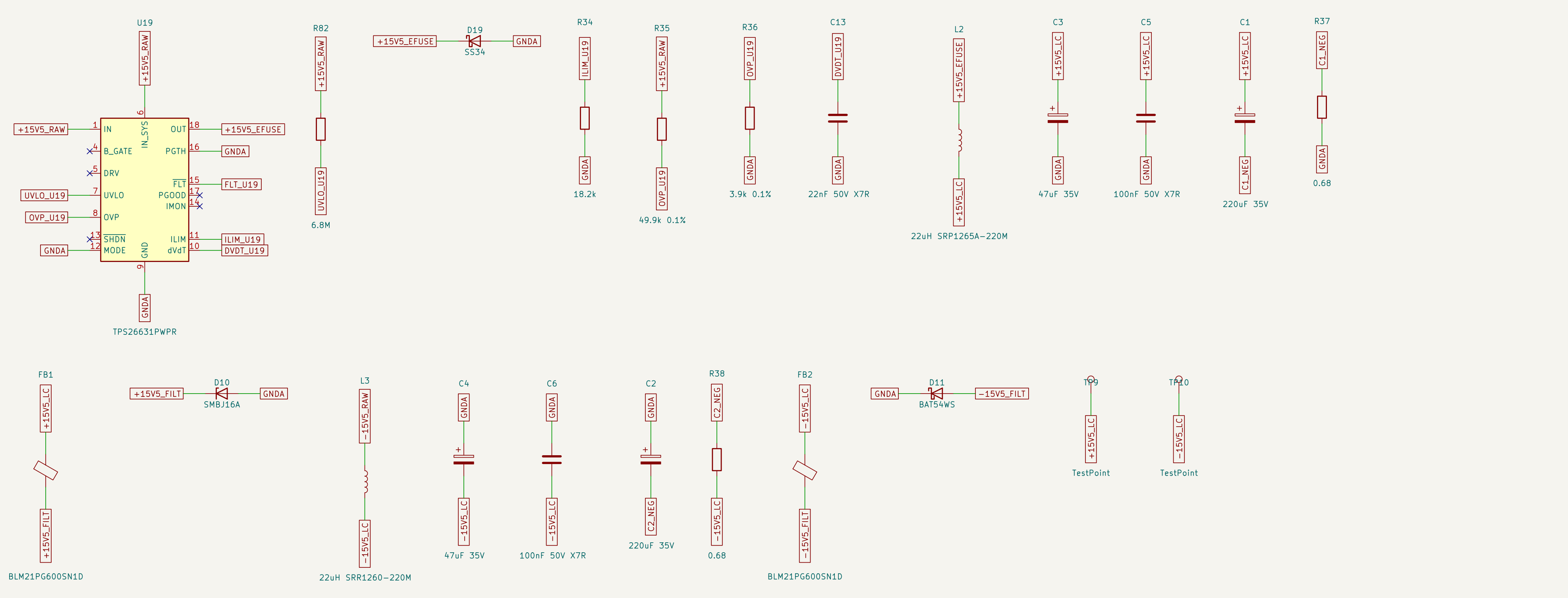
BLOQUE M5: Entrada 24 V, polaridad (LM74502 + CSD18534Q5A), eFuse U16 y supervisor M5


J1
Bornera 2P 5.0 mm
+24V_RAW
GND24
TP1
TP2
TestPoint
D1
SMBJ26CA
Q2
GATE_Q2
+24V_PROT
CSD18534Q5A
U15
LM74502DDFR
VS
EN/UVLO
OV
GND
GATE
SRC
VCAP
NC
C9
VCAP_U15
100nF 50V X7R
U16
R83
SHDN_M5
R11
ILIM_U16
R12
R13
UVLO_U16
R14
OVP_U16
C11
DVDT_U16
R64
D2
BZT52C24
+24V_LDO
C7
C8
4.7uF 25V X7R
MSP430FR2111IPW16
DVCC
DVSS
RST/NMI/SBWTDIO
TEST/SBWTCK
P2.7/XIN
P2.6/XOUT
+3V3_MCU
MCU_RST
MCU_TEST
FLT_M5
FLT_U19
LED_CTRL
SW_RESET
C23
100nF 50V X7R
R65
47k
C12
1nF 50V C0G
R66
10k
R84
10k
SW1
Pulsador 6x6
R31
LED_ANODE
220
J10
PinHeader 1x04

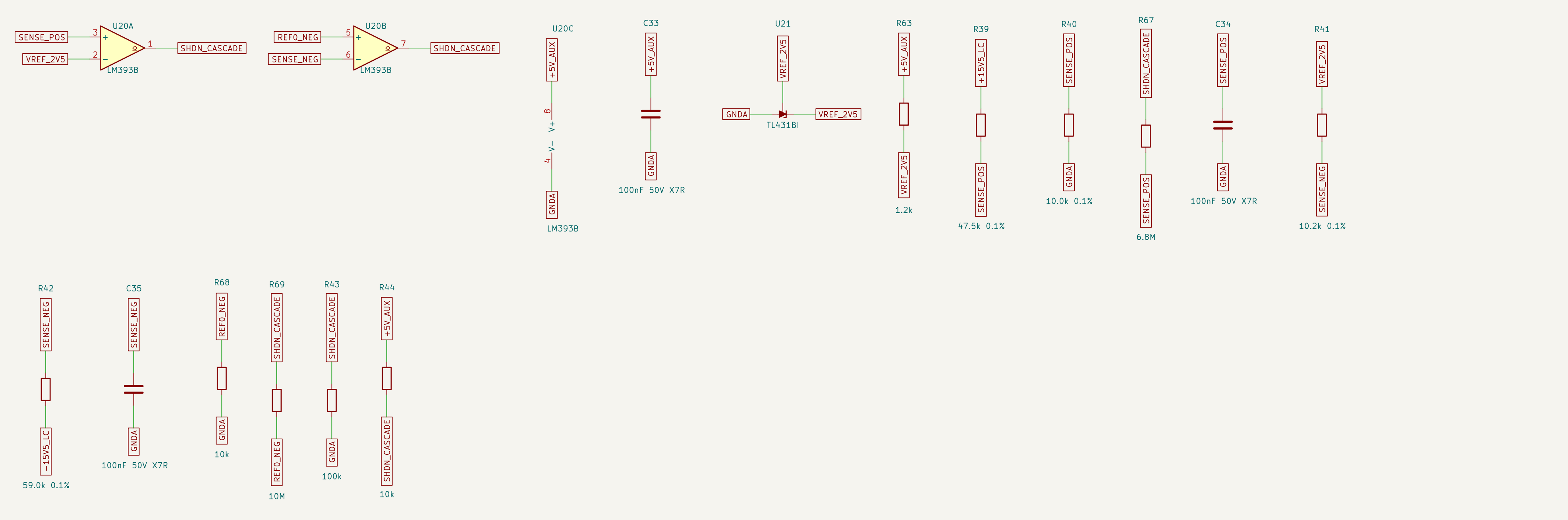
BLOQUE DCDC: Convertidores: THN 30 (+-12 rack), URB (+5 digital M1), K78 (+5 rack), 2 x THN 20 (+-15.5)


C14
100uF 50V
PS1
THN 30-2422WI
+24V_SW
GND24
+12V_RAW_RACK
GNDA
-12V_RAW_RACK
PS2
URB2405YMD-20WR3
+5V_RAW_DIG
GNDD
C29
100uF 25V
PS3
K7805T-1000R3
+5V_RAW_RACK
C15
10uF 50V X7R
PS4
THN 20-2413WI
+15V5_RAW
TRIM_PS4
R32
45.3k
PS5
THN 20-2413WI
TRIM_PS5
-15V5_RAW
R33
45.3k

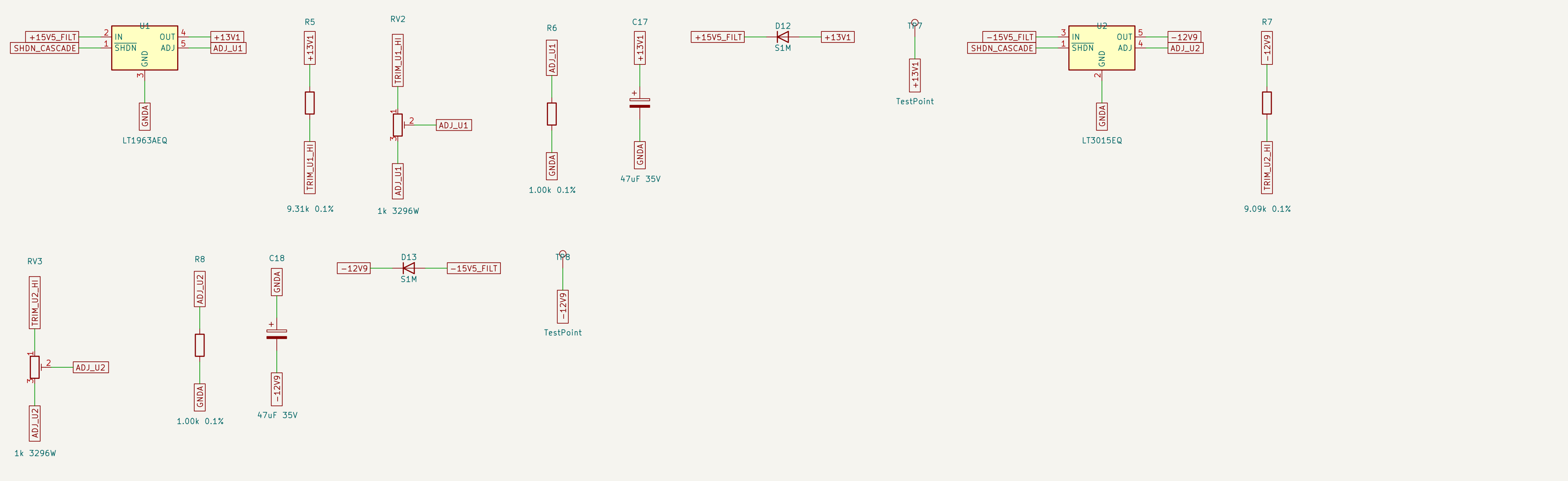
BLOQUE LC: Tope M3 (#334): TPS26631 + filtro LC 22 uH / 47 uF / 0.68 ohm + 220 uF + SMBJ16A


U19
TPS26631PWPR
+15V5_RAW
+15V5_EFUSE
R82
6.8M
D19
SS34
R34
18.2k
R35
49.9k 0.1%
R36
3.9k 0.1%
C13
22nF 50V X7R
L2
22uH SRP1265A-220M
C3
47uF 35V
FB1
BLM21PG600SN1D
D10
SMBJ16A
+15V5_FILT
L3
22uH SRR1260-220M
C4
47uF 35V
C6
100nF 50V X7R
C2
220uF 35V
R38
0.68
FB2
-15V5_LC
-15V5_FILT

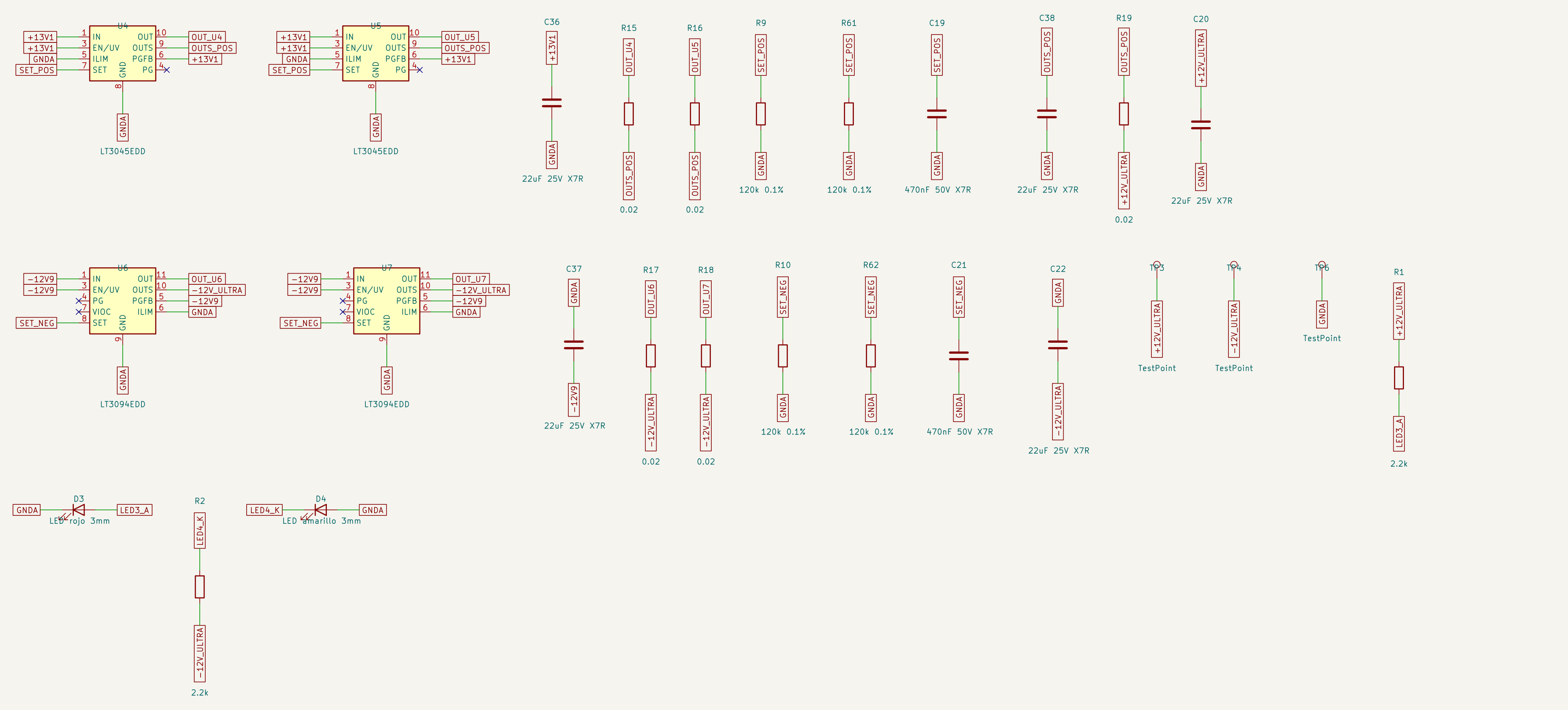
BLOQUE DF4: Detector D-F4: LM393 + TL431, umbral 14.5 V con histeresis


U20A
U20B
U20C
LM393B
SENSE_POS
VREF_2V5
REFO_NEG
SENSE_NEG
SHDN_CASCADE
C33
100nF 50V X7R
U21
TL431BI
R42
59.0k 0.1%
C35
100nF 50V X7R
R68
10k
R69
10M
R43
100k
R44
10k

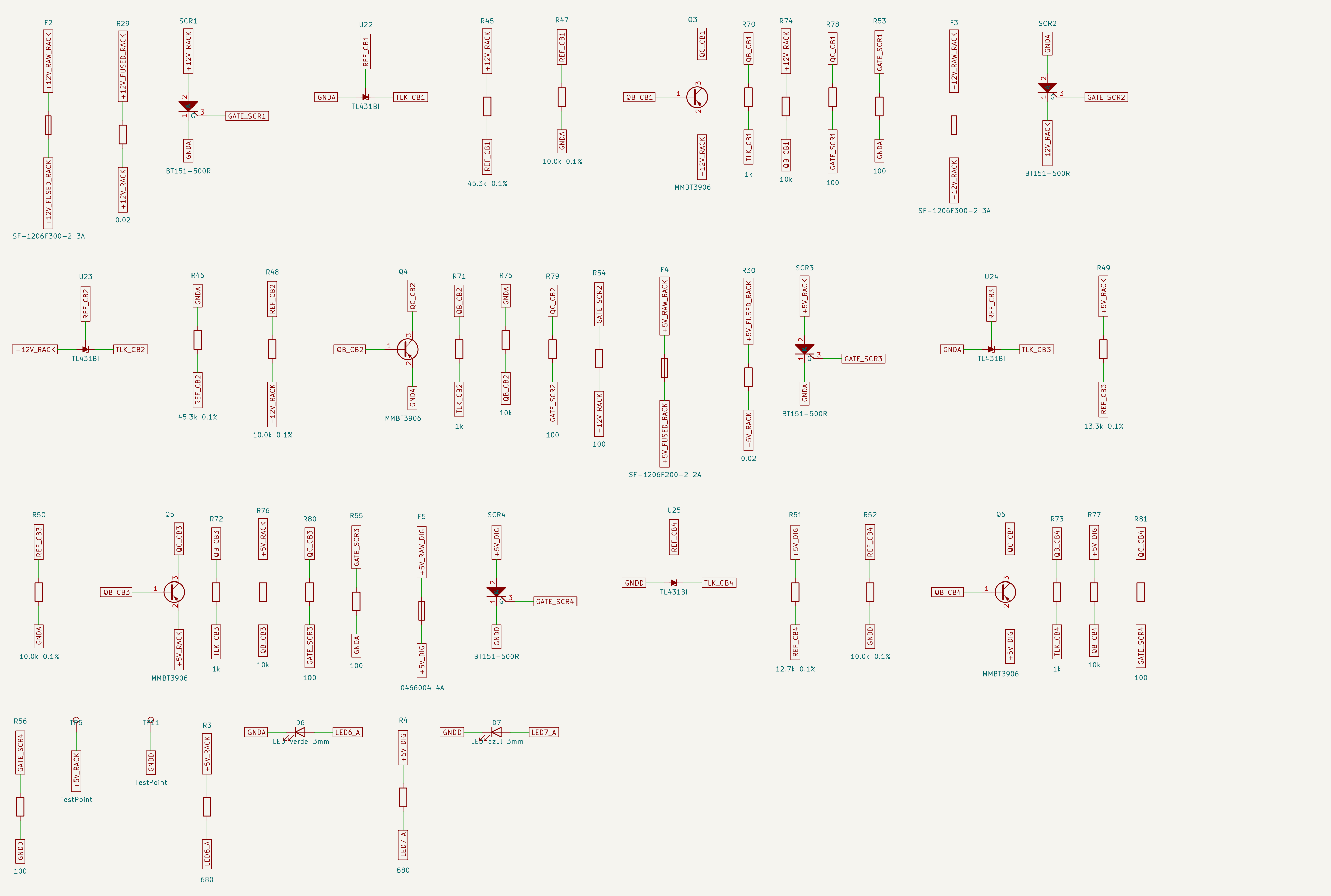
BLOQUE PRE: Pre-reguladores LT1963A (+13.1 V) y LT3015 (-12.9 V)


U1
LT1963AEQ
+15V5_FILT
SHDN_CASCADE
+13V1
ADJ_U1
R5
9.31k 0.1%
RV2
1k 3296W
R6
1.00k 0.1%
C17
47uF 35V
D12
S1M
RV3
1k 3296W
R8
1.00k 0.1%
C18
47uF 35V
D13
S1M
-12V9
-15V5_FILT
TP8
TestPoint

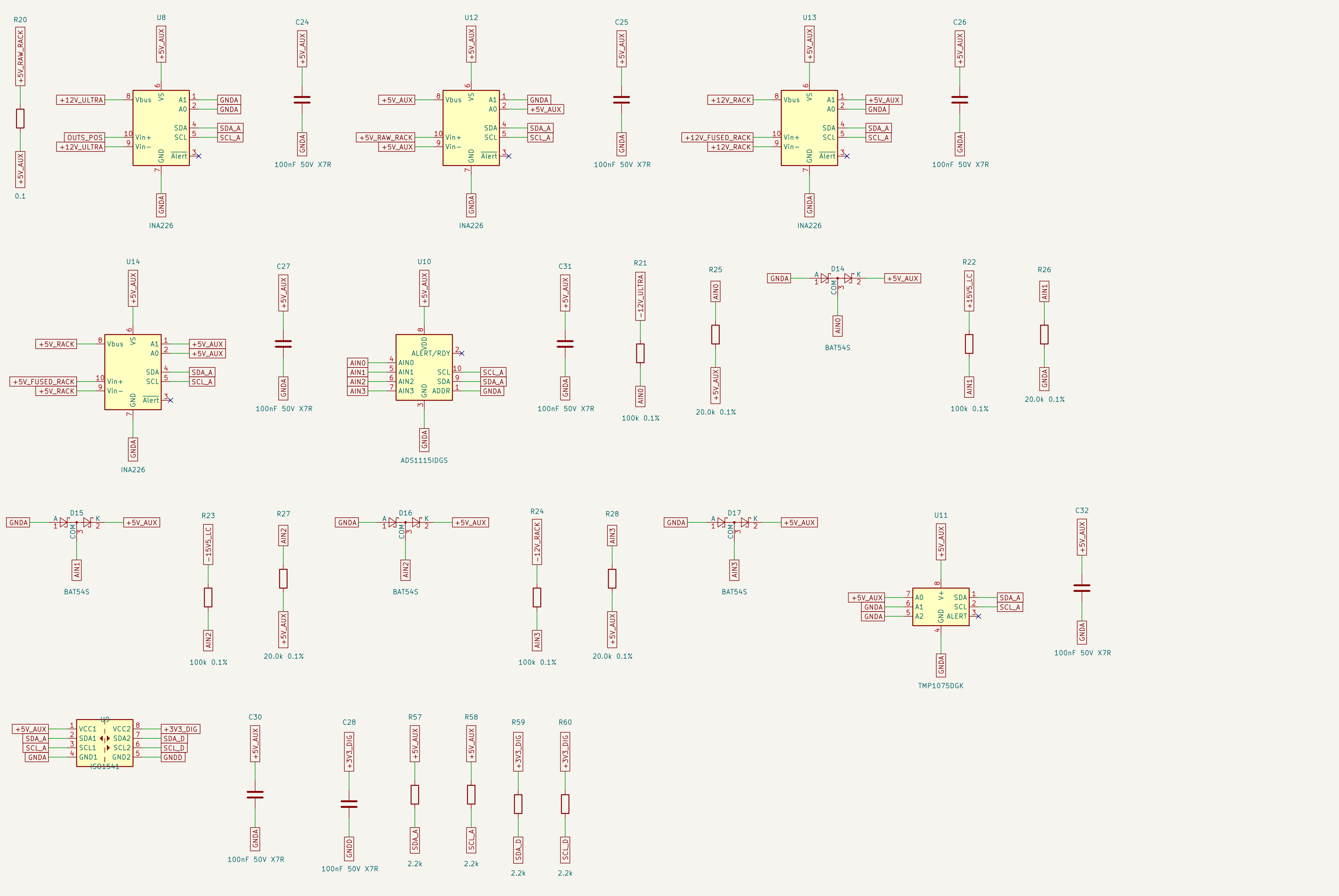
BLOQUE ULTRA: Cascada ultra: 2 x LT3045 y 2 x LT3094 en paralelo, R_SET = 60k


U4
U5
LT3045EDD
U6
U7
LT3094EDD
C36
22uF 25V X7R
R15
R16
0.02
R9
R61
120k 0.1%
C37
R17
R18
R10
R62
OUTS_POS
SET_POS
SET_NEG
-12V_ULTRA
D3
LED rojo 3mm
R2
2.2k
D4
LED amarillo 3mm

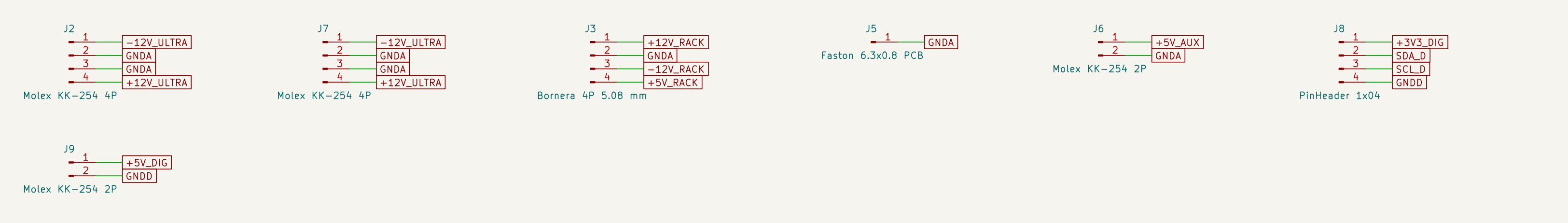
BLOQUE DF6: Rack: fusibles y crowbars D-F6 (TL431 -> PNP -> BT151)


F2
SF-1206F300-2 3A
R29
0.02
SCR1
BT151-500R
U22
TL431BI
R45
45.3k 0.1%
R47
10.0k 0.1%
Q3
MMBT3906
0466004 4A
R56
TP5
TP11
TestPoint
R3
680
D6
LED verde 3mm
R4
680
D7
LED azul 3mm

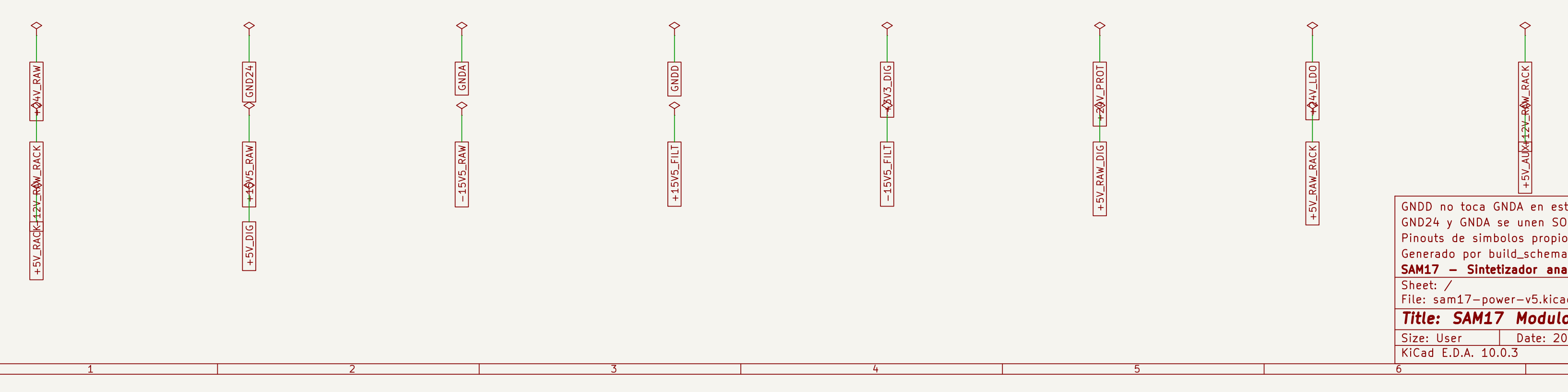
BLOQUE TEL: Telemetria I2C: 4 x INA226, ADS1115, TMP1075 e ISO1541


R20
0.1
U8
U12
INA226
C24
C25
100nF 50V X7R
U9
ISO1541
C30
C28
100nF 50V X7R
R57
R58
R59
R60
2.2k

BLOQUE CON: Conectores


J2
Molex KK-254 4P
J7
Molex KK-254 4P
J3
Bornera 4P 5.08 mm
J5
Faston 6.3x0.8 PCB
J9
Molex KK-254 2P
J8
PinHeader 1x04
-12V_ULTRA
GNDA
+12V_ULTRA
+12V_RACK
-12V_RACK
+5V_RACK
+5V_DIG
GNDD
+3V3_DIG
SDA_D
SCL_D

PWR_FLAG: redes que alimenta algo pasivo o un conector


+24V_RAW
+12V_RAW_RACK
+5V_RACK
GND24
+15V5_RAW
+5V_DIG
GNDA
-15V5_RAW
GNDD
+15V5_FILT
+3V3_DIG
-15V5_FILT
+5V_AUX
+12V_RAW_RACK

GNDD no toca GNDA en esta placa (M1): ISO1541 y URB aislados
GND24 y GNDA se unen SOLO en NT1, en los pads de masa de PS3 (K78 no aislado)
Pinouts de simbolos propios en build_simbolos.py, con su pagina de hoja
Generado por build_schematic.py: no editar a mano
SAM17 — Sintetizador analogico modular

Sheet: /
File: sam17-power-v5.kicad_sch

Title: SAM17 Modulo 01 v5 — Fuente: ladrillo 24 V y convertidores DC

Size: User | Date: 2026-10-22 | **Rev: v5-B**
KiCad E.D.A. 10.0.3 | Id: 1/1